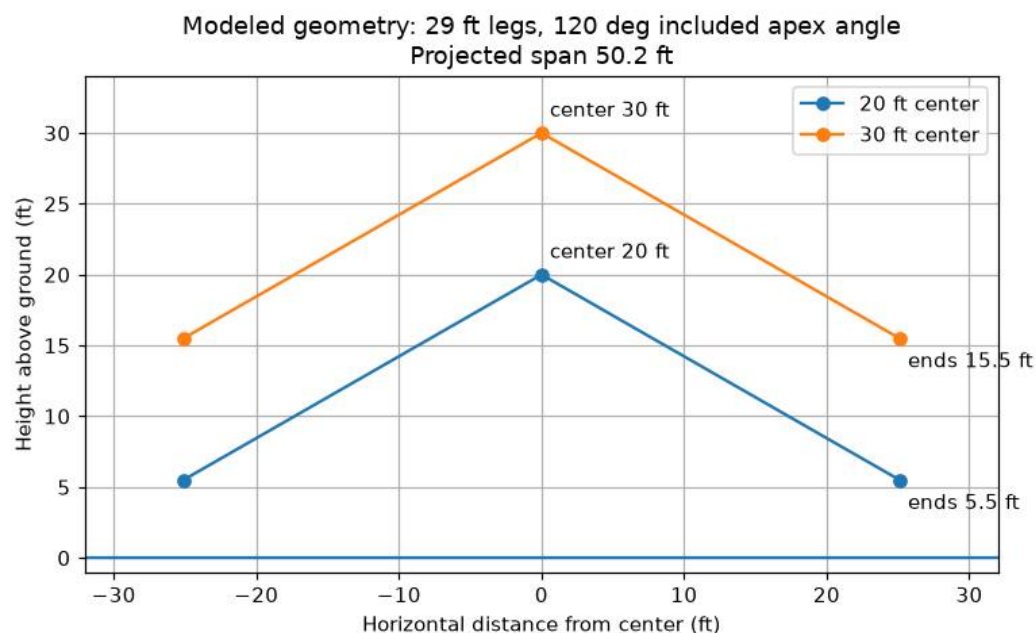




Geometry used

The user specified a 58 ft wire, 29 ft per leg, an apex wider than 90 degrees, and ends above 5 ft. A 120 degree included apex angle is used for both cases. This is essentially the minimum common symmetric geometry that keeps the ends above 5 ft with a 20 ft center. It yields a 50.2 ft projected span, 5.5 ft ends at the 20 ft center height, and 15.5 ft ends at the 30 ft center height.

Electrical assumptions

The wire is treated as a lossless thin conductor with the usual sinusoidal standing-current distribution. Its vector far field is integrated directly. Average ground is represented by complex TE/TM Fresnel reflection coefficients (relative permittivity 13, conductivity 0.005 S/m). The numerical line-current integrator was checked against the closed-form straight thin-dipole pattern; maximum normalized field amplitude error over the modeled electrical lengths was 2.02e-05.

Feed system assumption

The 28 ft homebrew balanced line is omitted from the far-field calculation. With equal and opposite differential currents on conductors spaced only 14 mm apart, its radiation should be small. The Mix 31 current choke at the KX2/feedline transition is therefore important: common-mode current would make the vertical feedline part of the antenna and could rotate or fill pattern nulls.

How to read the plots

Every pattern is normalized to its own strongest direction: 0 dB is the peak for that height and band. The plots show shape, bearing, and takeoff angle, not realized gain. The modeled wire runs east-west, so north-south is broadside. Rotate all compass bearings to match the actual deployed wire.

Pattern summary

Frequencies selected in the lower CW portions of each band

Band	L / lambda	20 ft center	30 ft center	Practical reading
40m 7.05 MHz	0.42	Peak 90 deg N; best 15 deg -9.5 dB N	Peak 90 deg N; best 15 deg -8.2 dB N	High-angle/NVIS at both heights.
20m 14.05 MHz	0.83	Peak 90 deg N; best 15 deg -7.4 dB N	Peak 41 deg N; best 15 deg -4.3 dB N	30 ft produces a distinct mid-angle broadside lobe.
17m 18.08 MHz	1.07	Peak 68 deg N; best 15 deg -6.6 dB N	Peak 34 deg N; best 15 deg -3.2 dB N	30 ft is substantially better for lower-angle broadside work.
15m 21.05 MHz	1.24	Peak 90 deg N; best 15 deg -7.4 dB N	Peak 33 deg N; best 15 deg -3.1 dB N	30 ft gives the cleanest broadside EDZ-like pattern.
12m 24.91 MHz	1.47	Peak 90 deg N; best 15 deg -6.9 dB NE	Peak 41 deg N; best 15 deg -3.6 dB NE	Extra lobes begin; low-angle diagonals become important.
10m 28.05 MHz	1.65	Peak 90 deg N; best 15 deg -4.5 dB NE	Peak 46 deg N; best 15 deg -0.4 dB NE	Strongly multi-lobed; diagonal low-angle lobes dominate DX bearings.

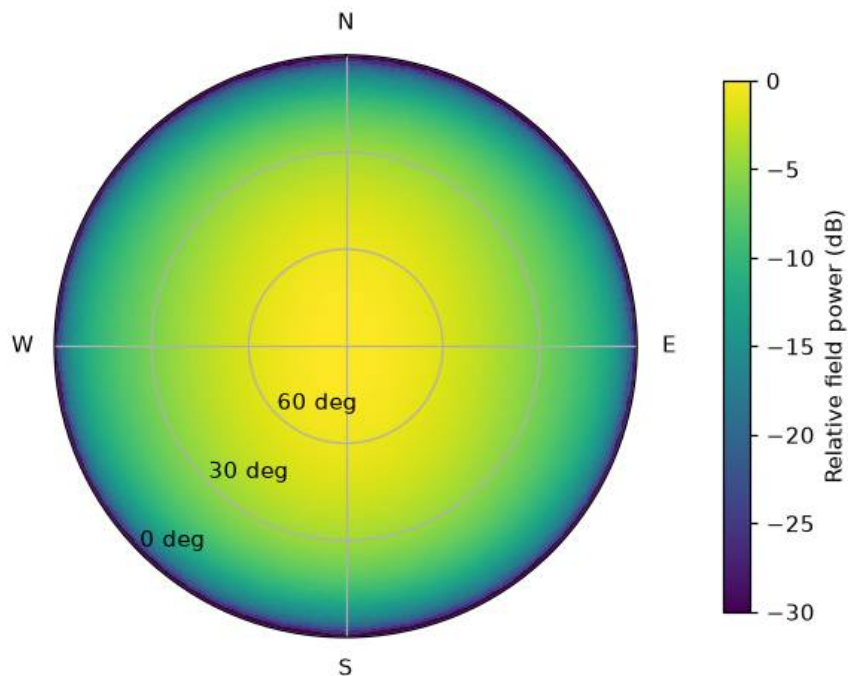
Interpretation: the 15 degree figures are relative to the overall peak for that same configuration, not relative to the best direction within the 15 degree cut. Symmetry creates equivalent mirror lobes, so a single compass label in the table represents a family of equivalent bearings.

40m pattern - 7.05 MHz

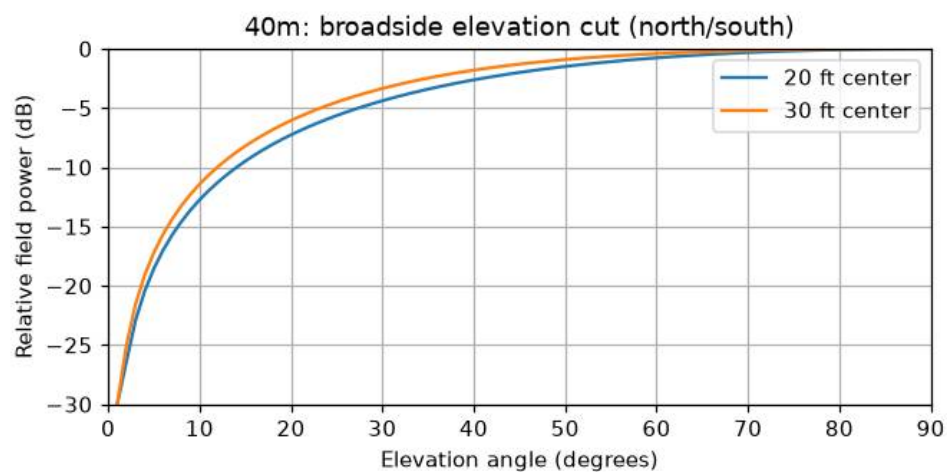
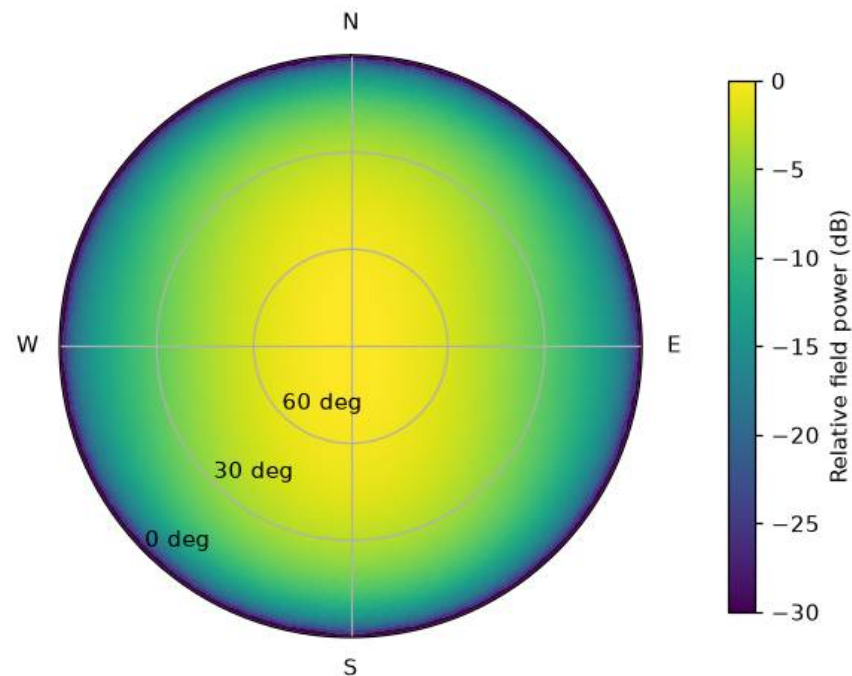
58 ft total = 0.42 wavelengths

Sky views: center = zenith, rim = horizon. All scales are dB relative to that configuration's own peak.

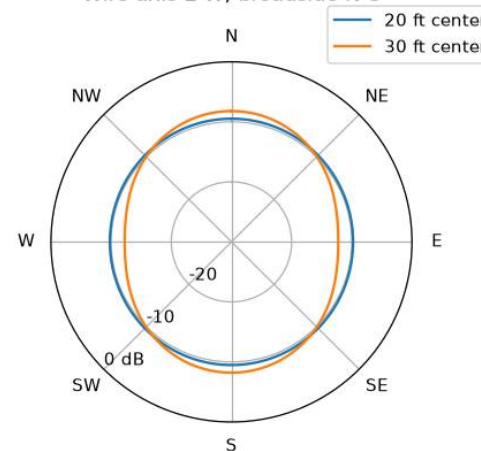
40m - 7.05 MHz - 20 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



40m - 7.05 MHz - 30 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



40m: azimuth cut at 15 deg elevation
Wire axis E-W; broadside N-S



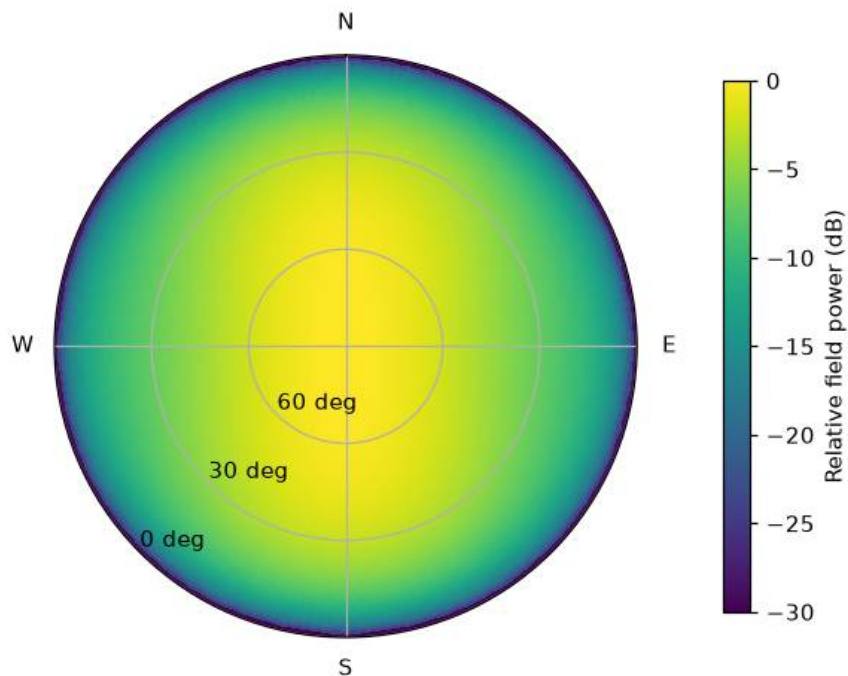
Result: Both installations are dominated by a zenith lobe: this is an NVIS/regional pattern. Raising the center from 20 to 30 ft improves the best 15 deg radiation by only about 1.3 dB.

20m pattern - 14.05 MHz

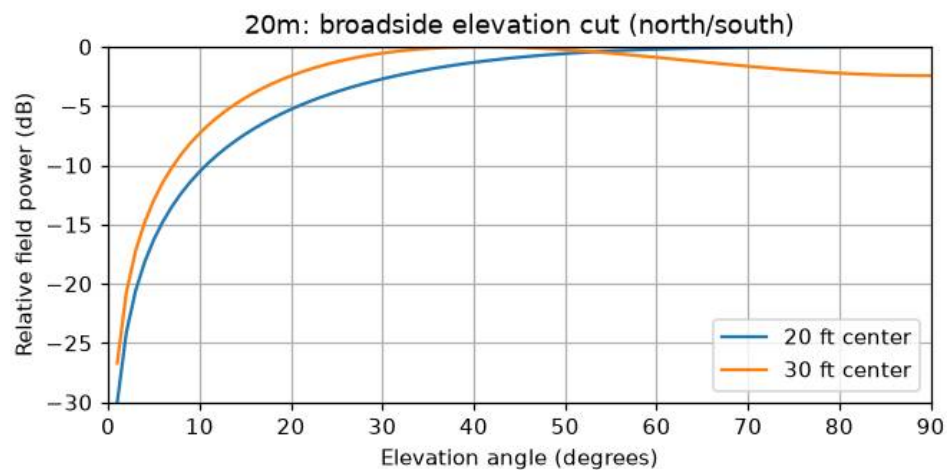
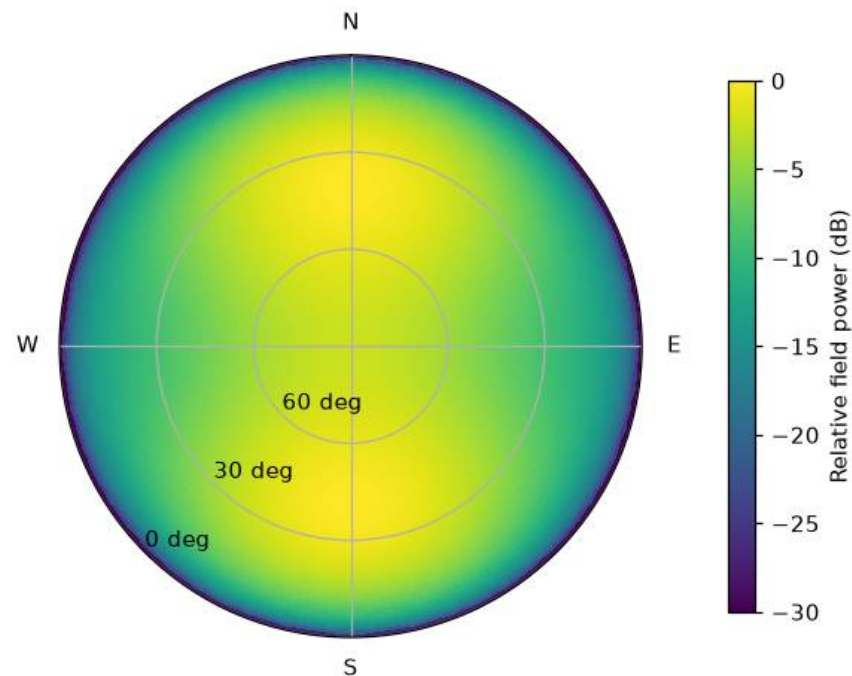
58 ft total = 0.83 wavelengths

Sky views: center = zenith, rim = horizon. All scales are dB relative to that configuration's own peak.

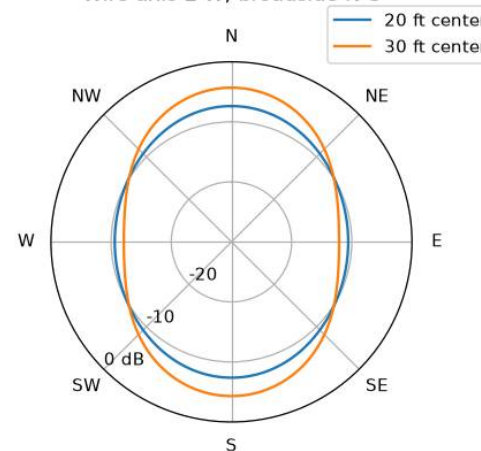
20m - 14.05 MHz - 20 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



20m - 14.05 MHz - 30 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



20m: azimuth cut at 15 deg elevation
Wire axis E-W; broadside N-S



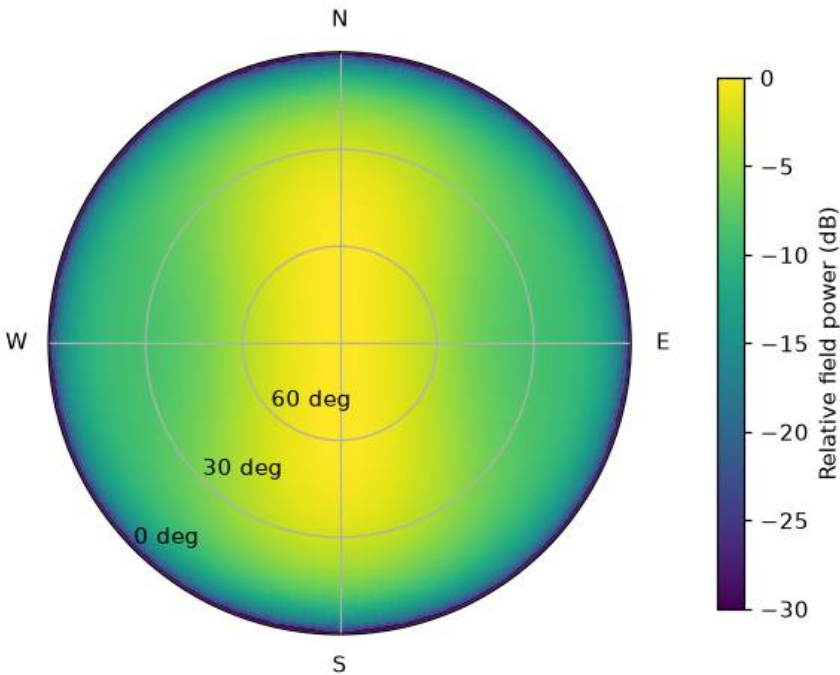
Result: At 20 ft the strongest lobe remains vertical. At 30 ft the main broadside lobe moves to about 41 deg, and the best 15 deg radiation improves by about 3.1 dB.

17m pattern - 18.08 MHz

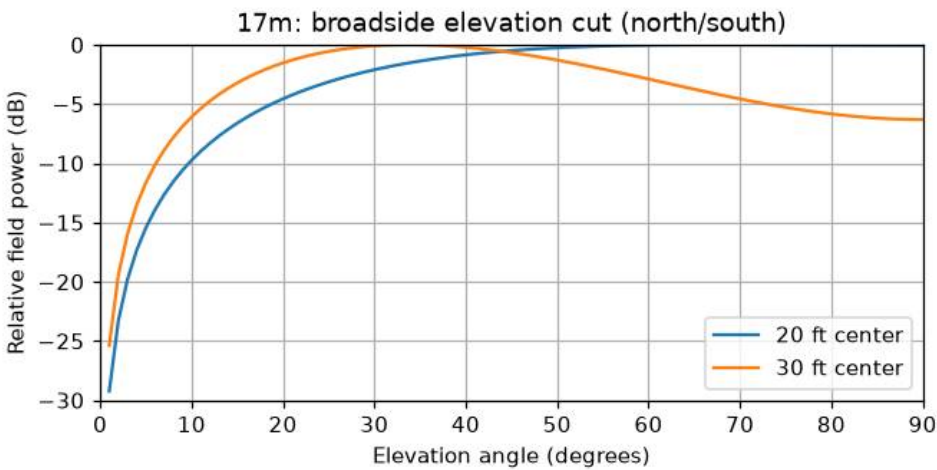
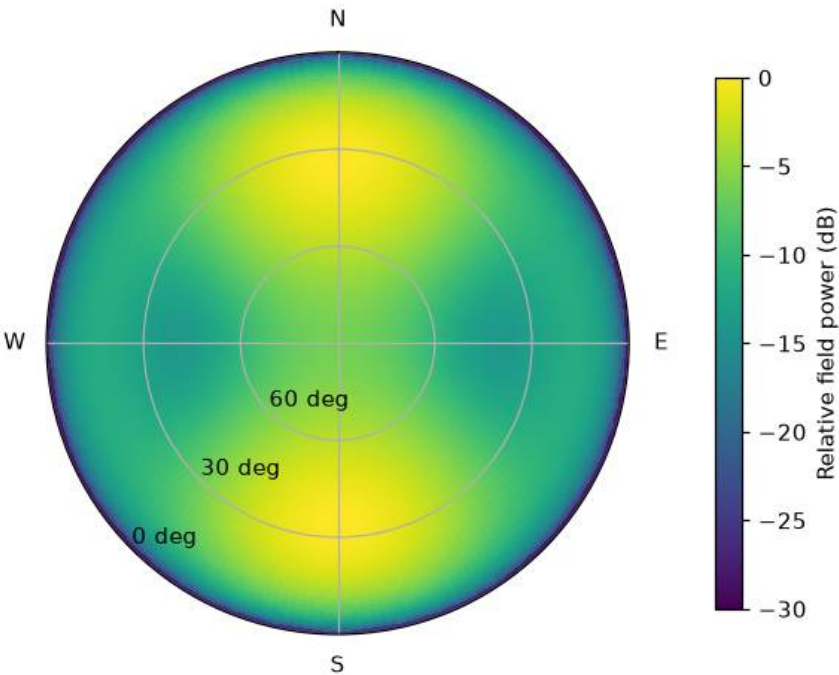
58 ft total = 1.07 wavelengths

Sky views: center = zenith, rim = horizon. All scales are dB relative to that configuration's own peak.

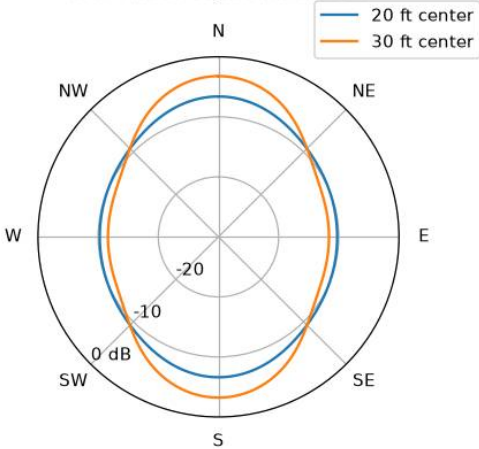
17m - 18.08 MHz - 20 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



17m - 18.08 MHz - 30 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



17m: azimuth cut at 15 deg elevation
Wire axis E-W; broadside N-S



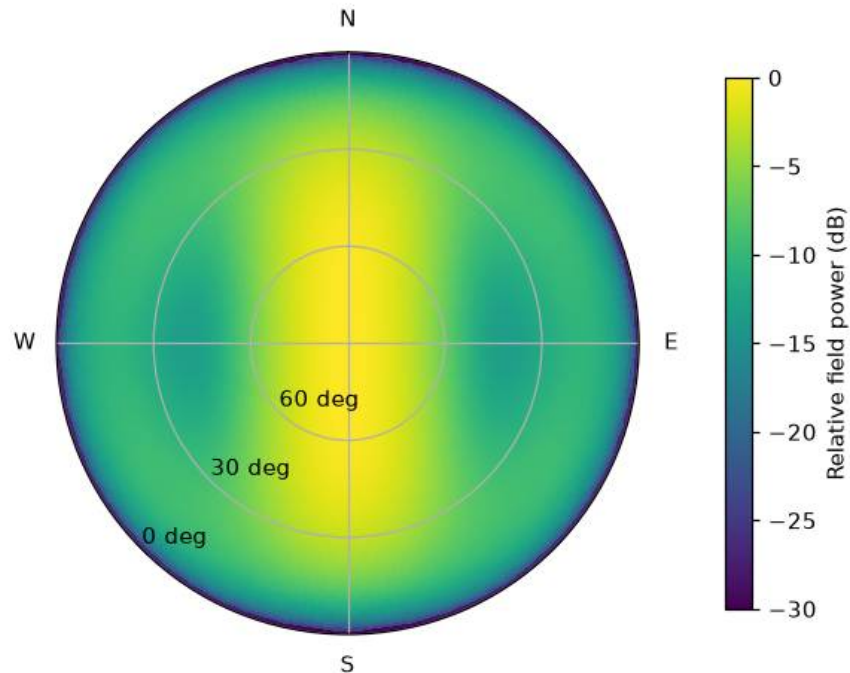
Result: The 20 ft installation remains high-angle, peaking near 68 deg. At 30 ft the broadside main lobe drops to about 34 deg.

15m pattern - 21.05 MHz

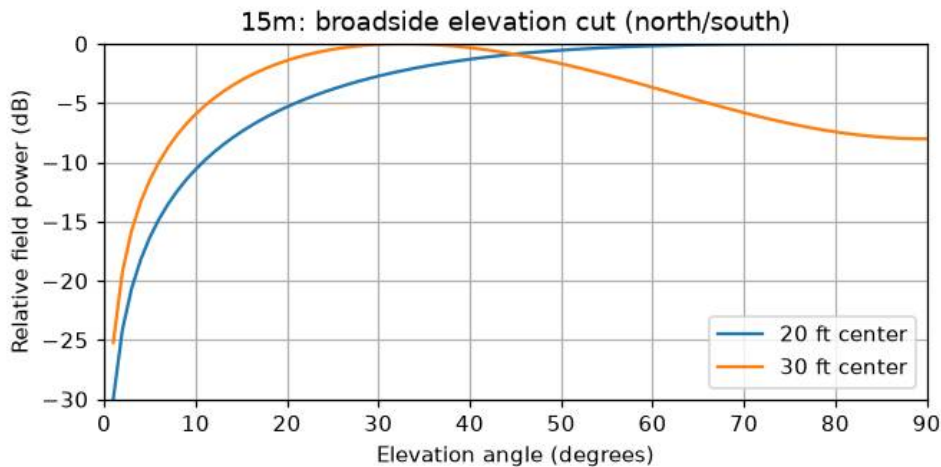
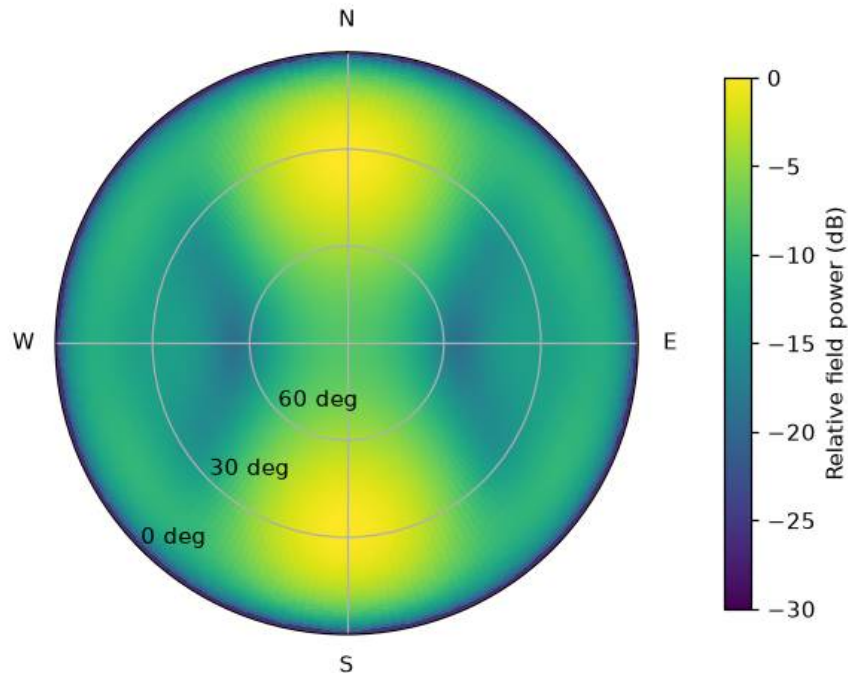
58 ft total = 1.24 wavelengths

Sky views: center = zenith, rim = horizon. All scales are dB relative to that configuration's own peak.

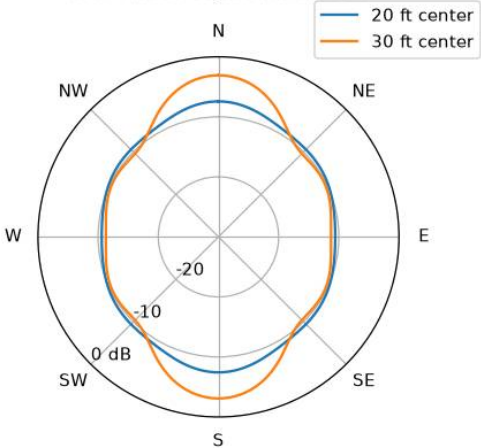
15m - 21.05 MHz - 20 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



15m - 21.05 MHz - 30 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



15m: azimuth cut at 15 deg elevation
Wire axis E-W; broadside N-S

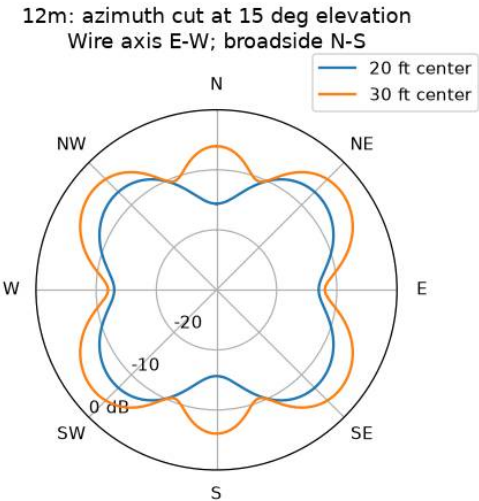
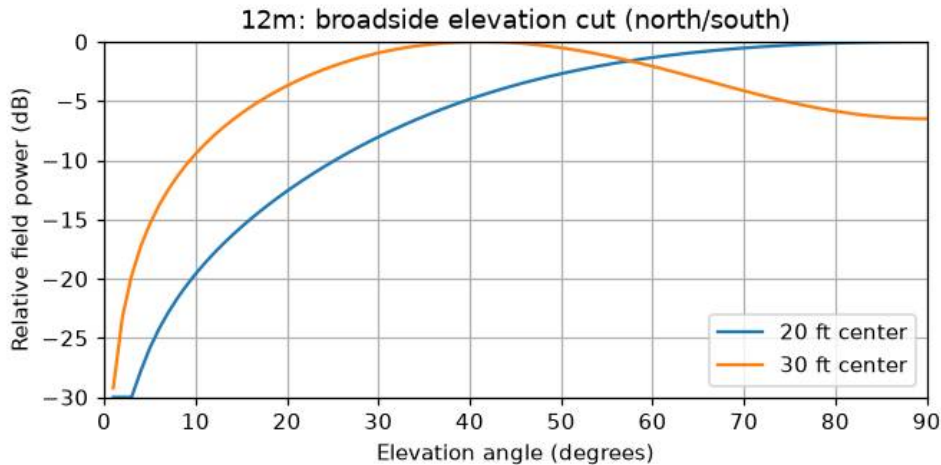
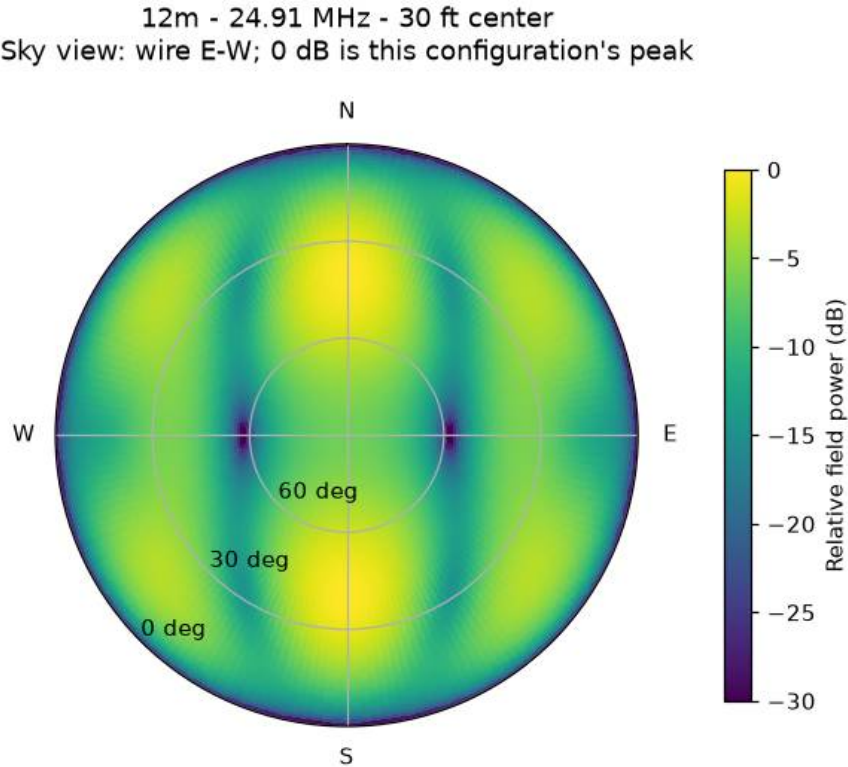
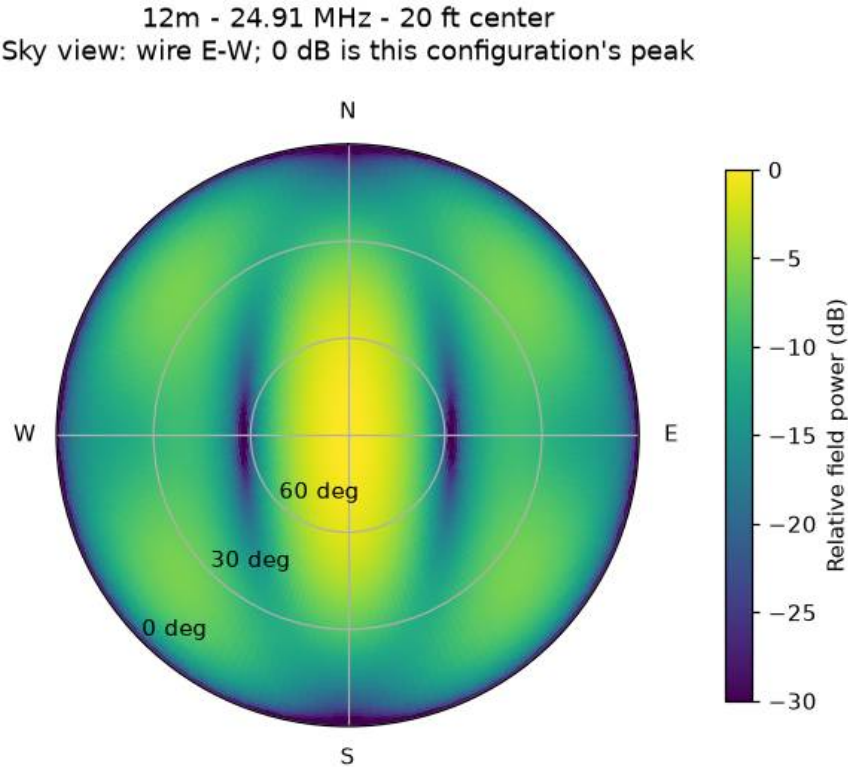


Result: At 30 ft this is the cleanest extended-double-Zepp-like result: a broadside main lobe near 33 deg. At 20 ft, ground interaction pulls the strongest energy back toward the zenith.

12m pattern - 24.91 MHz

58 ft total = 1.47 wavelengths

Sky views: center = zenith, rim = horizon. All scales are dB relative to that configuration's own peak.



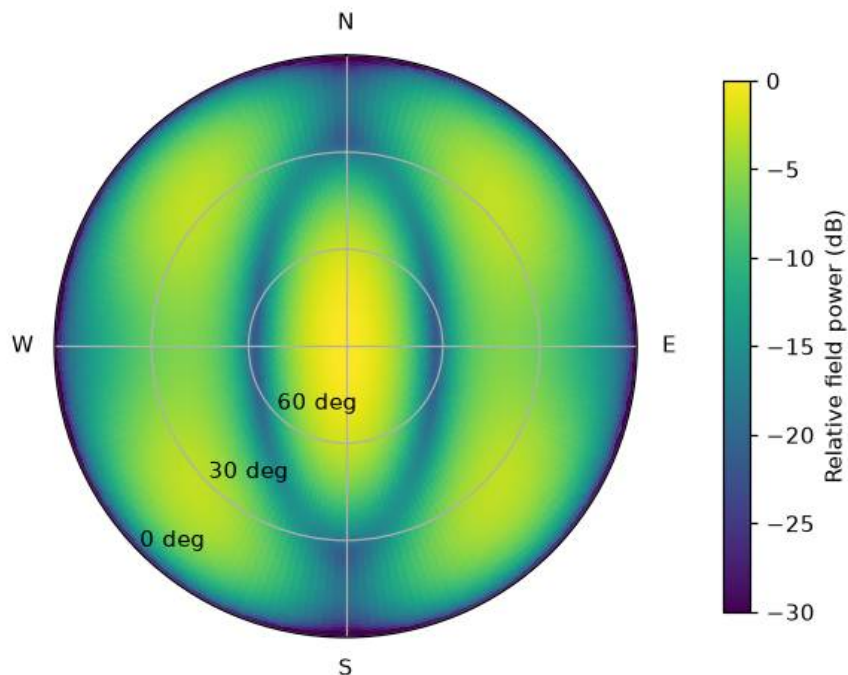
Result: Additional azimuth lobes are now visible. The 30 ft installation still has a broadside main lobe near 41 deg, but at 15 deg elevation the best bearings are diagonal rather than pure broadside.

10m pattern - 28.05 MHz

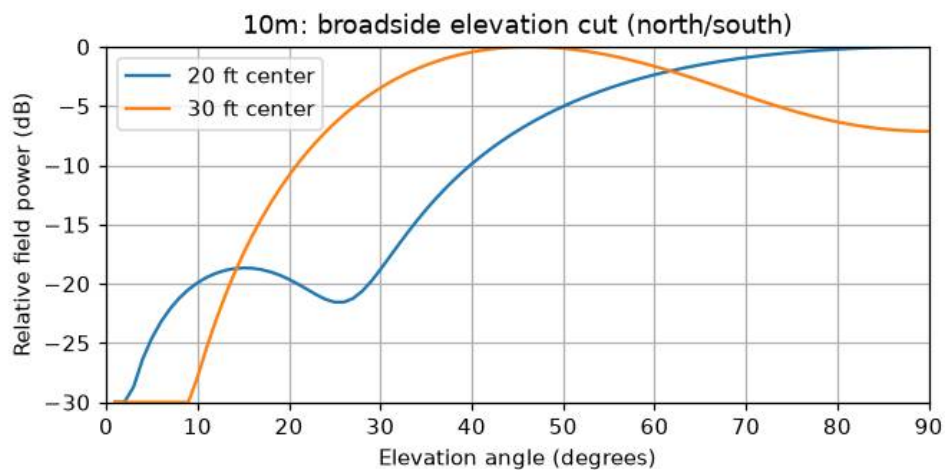
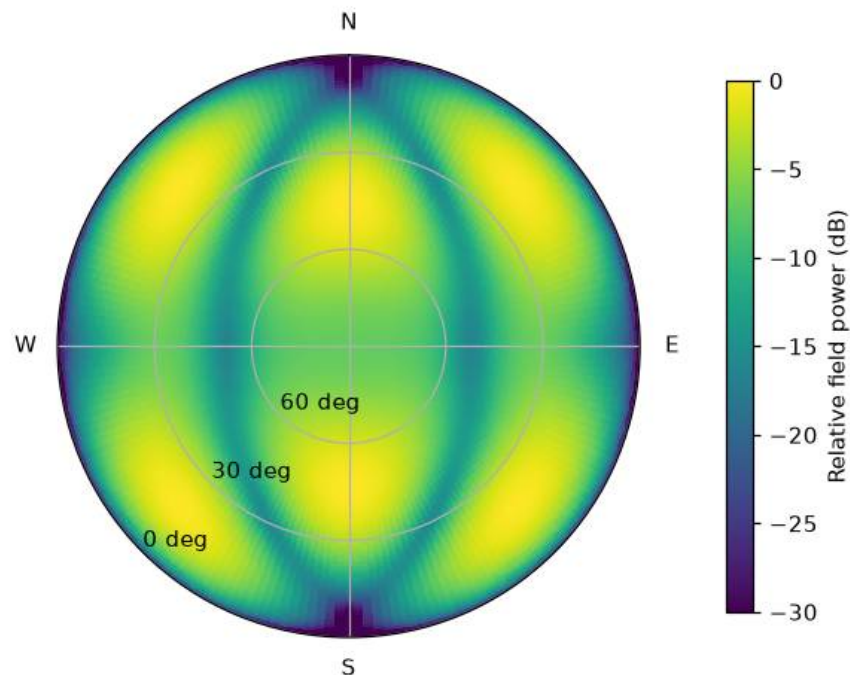
58 ft total = 1.65 wavelengths

Sky views: center = zenith, rim = horizon. All scales are dB relative to that configuration's own peak.

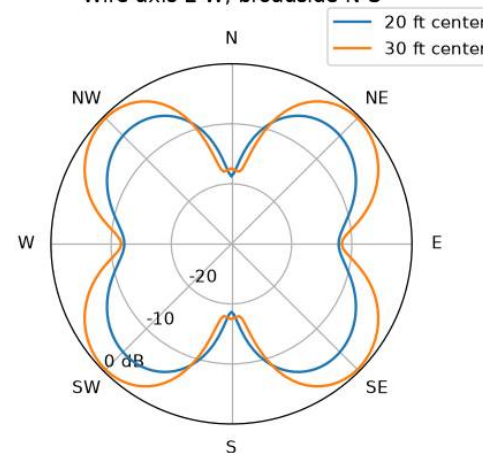
10m - 28.05 MHz - 20 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



10m - 28.05 MHz - 30 ft center
Sky view: wire E-W; 0 dB is this configuration's peak



10m: azimuth cut at 15 deg elevation
Wire axis E-W; broadside N-S



Result: This band is decisively multi-lobed. At 30 ft, low-angle diagonal lobes peak near 18 deg and are only 0.1 dB below the overall peak; pure broadside is close to a low-angle null.

40 m: Both heights are primarily high-angle. The extra 10 ft does not turn this into a low-angle DX antenna.

20/17/15 m: The 30 ft center is materially better. Main broadside lobes are approximately 41, 34, and 33 degrees respectively. At 20 ft, ground interaction keeps the strongest radiation much higher.

12 m: The antenna is now about 1.47 wavelengths long. It remains usable, but azimuth lobes are developing and low-angle best bearings are no longer simply broadside.

10 m: Expect four useful diagonal low-angle lobes rather than a classic two-lobe dipole pattern. With the wire east-west, those lobes are approximately toward the northeast, southeast, southwest, and northwest.

Feedline: The pattern assumes the two-wire line remains balanced. If the Mix 31 choke is ineffective at a particular band, or the two legs are deployed asymmetrically, common-mode current can substantially alter the result.

Uncertainty: Real soil, sloping terrain, nearby trees, wire sag, and endpoint placement can move peak angles by several degrees and fill deep nulls. The broad pattern transitions are more trustworthy than the exact depth of any narrow null.

30° elevation resolution - normalized to the wire's radiation pattern
Minimum gain: 0 dB (0.0001 W/m²) Maximum gain: 30 dB (0.0001 W/m²) Center of peak is at 0.0001 W/m²

